

**DESIGN AND IMPLEMENTATION OF AN
ARITHMETIC LOGIC UNIT (ALU)
USING VERILOG HDL**

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Digital System Design Laboratory Report

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1. Abstract

This report presents the comprehensive design and simulation of an Arithmetic Logic Unit (ALU) using Verilog Hardware Description Language (HDL). An ALU is a fundamental digital circuit that forms the computational core of every processor and microcontroller. The ALU described in this report is capable of performing multiple arithmetic and logical operations — including addition, subtraction, multiplication, division, increment, decrement, bitwise operations (AND, OR, XOR, NOT), and shift operations (left shift and right shift).

The design is written in synthesizable Verilog code and follows standard combinational logic design principles. A dedicated Verilog testbench is used to drive input stimuli and verify the correctness of each operation. Simulation is performed to generate waveforms that visually confirm functional correctness under different operation selections. This report includes the full HDL source code, a code walkthrough, RTL schematic analysis, and simulation results. The design is suitable for FPGA implementation and academic study in digital systems engineering.

2. Introduction

An Arithmetic Logic Unit (ALU) is one of the most critical components of any digital processing system. It is a combinational circuit that accepts two operand inputs and a control signal (opcode), and produces an output based on the selected operation. The ALU serves as the heart of the Central Processing Unit (CPU), performing the low-level computation required to execute every instruction in a program.

In modern processors, the ALU is responsible not only for basic arithmetic like addition and subtraction, but also for logical comparisons, bit shifting, and other operations essential to program execution. Without the ALU, no computation can take place, which makes understanding its design fundamental to digital electronics and computer architecture education.

In this project, Verilog HDL is used to model an ALU at the Register Transfer Level (RTL). Verilog is an industry-standard hardware description language widely used in FPGA and ASIC design. The design uses a case statement inside an always block to map each opcode value to its corresponding operation. The result output is a 10-bit register that accommodates the wider range of values produced by multiplication and shift operations.

The design is verified using a testbench that applies fixed input values and cycles through every supported opcode. The simulation can be run using industry-standard tools such as Xilinx Vivado or Mentor ModelSim, both of which are described in the simulation section.

3. Objectives

The primary objectives of this project are as follows:

- To design a synthesizable and functionally correct ALU module using Verilog HDL that adheres to standard digital design practices.
- To implement a wide range of arithmetic operations (addition, subtraction, multiplication, division, increment, decrement) and logical operations (AND, OR, XOR, NOT) within a single unified module.

- To incorporate shift operations (left shift and right shift) that are critical for efficient hardware arithmetic and data manipulation.
- To implement opcode-based operation selection using a 4-bit control signal, allowing up to 16 unique operations to be encoded.
- To write a comprehensive testbench in Verilog that applies input stimuli and verifies the correctness of the ALU output for all operations.
- To simulate the design using EDA tools (Vivado and ModelSim) and analyze the resulting waveforms to confirm proper functionality.
- To produce an RTL schematic after synthesis to visualize the underlying gate-level hardware structure of the ALU.
- To evaluate the design's suitability for FPGA deployment by analyzing its combinational logic structure and resource utilization.

4. Design Methodology

The ALU is designed using combinational logic principles in Verilog HDL. The overall design methodology follows a top-down approach: the behavioral specification is first written in Verilog, then simulated to verify correctness, and finally synthesized to produce an RTL schematic.

4.1 Module Architecture

The ALU is encapsulated in a single Verilog module named `alu`. It takes three inputs: two 5-bit operands (`a` and `b`) and a 4-bit opcode (`op`). The output is a 10-bit register (`result`) that is wide enough to hold the maximum value produced by any operation, including multiplication of two 5-bit numbers (which can produce up to a 10-bit result).

4.2 Opcode-Based Control Logic

A 4-bit opcode provides 16 possible operation codes (0000 to 1111). This project uses 12 of those codes. Each code is mapped to a named local parameter (`localparam`) such as `add`, `sub`, `mul`, and so on. This makes the code highly readable and maintainable. The case statement inside the `always` block evaluates the opcode on every input change and routes the computation accordingly.

4.3 Combinational Logic with `always @(*)`

The `always @(*)` sensitivity list means the `always` block triggers whenever any of the input signals (`a`, `b`, or `op`) change. This makes the ALU a purely combinational circuit — there are no clock signals or sequential elements. The result updates instantaneously whenever the inputs change, which is the expected behavior for a combinational ALU.

4.4 Supported Operations

The following table summarizes all supported operations and their corresponding opcodes:

Opcode	Operation	Description
0000	add	Adds operands a and b
0001	sub	Subtracts b from a

0010	mul	Multiplies a and b
0011	div	Divides a by b (integer)
0100	incr	Increments a by 1
0101	decr	Decrements a by 1
0110	AND	Bitwise AND of a and b
0111	OR	Bitwise OR of a and b
1000	NOT	Bitwise NOT of a
1001	XOR	Bitwise XOR of a and b
1010	bitshift_left	Logical left shift of a by b bits
1011	bitshift_right	Logical right shift of a by b bits

5. Program Code & Dynamic Explanation

5.1 Verilog Design Code

The following is the complete Verilog HDL source code for the ALU module:

```

module alu(
    input  [4:0] a, b,
    input  [3:0] op,
    output reg [9:0] result
);

localparam add      = 4'b0000,
            sub      = 4'b0001,
            mul      = 4'b0010,
            div      = 4'b0011,
            incr     = 4'b0100,
            decr     = 4'b0101,
            AND      = 4'b0110,
            OR       = 4'b0111,
            NOT      = 4'b1000,
            XOR      = 4'b1001,
            bitshift_left = 4'b1010,
            bitshift_right= 4'b1011;

initial result = 0;

always @(*) begin
    case(op)
        add:      result = a + b;
        sub:      result = a - b;
        mul:      result = a * b;
        div:      result = a / b;
        incr:     result = a + 1;
    endcase
end

```

```

    decre:      result = a - 1;
    AND:        result = a & b;
    OR:         result = a | b;
    NOT:        result = ~a;
    XOR:        result = a ^ b;
    bitshift_left: result = a << b;
    bitshift_right: result = a >> b;
endcase
end
endmodule

```

5.2 Code Component Breakdown and Analysis

Each section of the Verilog code plays a specific role in the ALU design. Below is a detailed explanation of every component:

Module Declaration: The module keyword defines the boundary of the hardware block. The port list specifies the interface: a and b are 5-bit wide input operands (allowing values from 0 to 31), op is a 4-bit opcode (allowing 16 unique operation codes), and result is a 10-bit output register. The output is declared as reg because it is driven inside an always block.

Local Parameters (localparam): The localparam declarations assign human-readable names to the 4-bit binary opcode values. Using named constants instead of raw binary literals makes the code self-documenting and greatly reduces the chance of coding errors. These constants are synthesized away by the compiler and have no hardware footprint.

Initial Block: The statement initial result = 0 sets the initial value of the result register to zero when the simulation begins. In real hardware (FPGA), this corresponds to the power-on reset value of the register.

always @(*) Block: This is the core of the ALU. The sensitivity list @(*) is an auto-sensitivity list that automatically includes all signals read inside the block — in this case, a, b, and op. Whenever any of these signals change, the always block re-evaluates. This ensures that the output always reflects the latest input values, which is the defining characteristic of a combinational circuit.

case Statement: Inside the always block, the case statement compares the value of op against each localparam value. When a match is found, the corresponding arithmetic or logical expression is evaluated and assigned to result. This is equivalent to a hardware multiplexer tree that routes the correct computational output to the result port.

Arithmetic Operations: The add operation uses Verilog's built-in + operator; sub uses -; mul uses *; div uses /. For integer division, the result is truncated. The incre and decre operations add or subtract the constant 1 from operand a, ignoring b entirely.

Logical Operations: AND uses &, OR uses |, XOR uses ^, and NOT uses ~ — all bitwise operators applied across all 5 bits simultaneously. The NOT operation is a unary operation (applies only to a), while the others are binary.

Shift Operations: bitshift_left uses the << operator to shift the bits of a to the left by b positions. Each left shift by 1 is equivalent to multiplying by 2. bitshift_right uses >> to shift bits rightward, equivalent to integer division by 2 per position. The 10-bit output width accommodates the expanded bit-width produced by shift operations.

5.3 Verification Testbench Code

The testbench instantiates the ALU module and applies a sequence of input stimuli to test every opcode. The testbench uses reg types for inputs and wire for outputs, which is the standard pattern for Verilog testbenches.

```
module tb_alu;
    reg [4:0] a, b;
    reg [3:0] op;
    wire [9:0] result;

    alu uut(a, b, op, result);

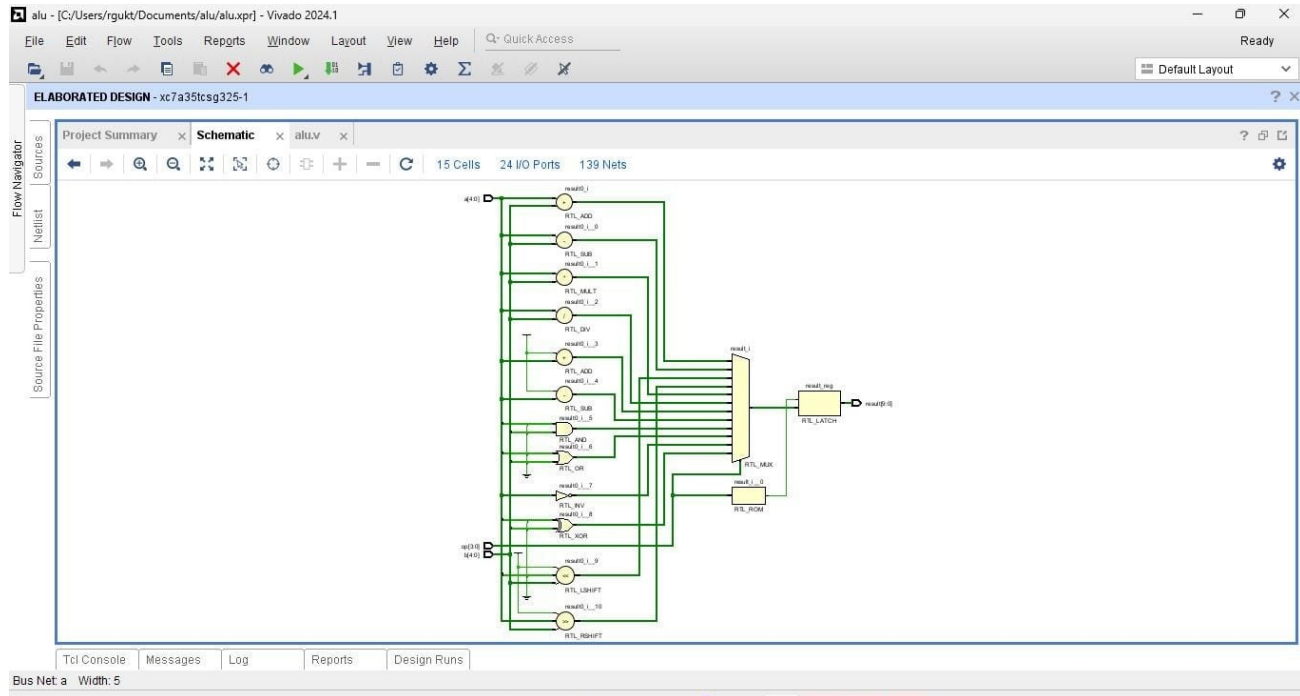
    localparam add      = 4'b0000,
               sub      = 4'b0001,
               mul      = 4'b0010,
               div      = 4'b0011,
               incre     = 4'b0100,
               decre     = 4'b0101,
               AND       = 4'b0110,
               OR        = 4'b0111,
               NOT       = 4'b1000,
               XOR       = 4'b1001,
               bitshift_left = 4'b1010,
               bitshift_right= 4'b1011;

    initial begin
        a = 5'd5; b = 5'd4;
        op = add;      #10;    // result = 9
        op = sub;      #10;    // result = 1
        op = mul;      #10;    // result = 20
        op = div;      #10;    // result = 1
        op = incre;    #10;    // result = 6
        op = decre;    #10;    // result = 4
        op = AND;      #10;    // result = 4
        op = OR;       #10;    // result = 5
        op = NOT;      #10;    // result = ~5
        op = XOR;      #10;    // result = 1
        op = bitshift_left; #10; // result = 5 << 4 = 80
        op = bitshift_right; #10; // result = 5 >> 4 = 0
        $finish;
    end
endmodule
```

Testbench Explanation: The testbench sets a = 5 (binary 00101) and b = 4 (binary 00100) as fixed operands, then cycles through all 12 opcodes with a 10-time-unit (#10) delay between each change. This delay gives the simulator time to propagate the combinational logic and capture stable output values. The \$finish system task terminates the simulation once all test cases have been applied.

Expected Results: For a=5 and b=4: ADD=9, SUB=1, MUL=20, DIV=1 (integer), INCRE=6, DECRE=4, AND=4, OR=5, NOT=111111111111010 (inverted 5 in 10 bits), XOR=1, LEFT SHIFT by 4 = 80, RIGHT SHIFT by 4 = 0. These can be verified against the simulation waveforms.

6. RTL Schematic Layout



After synthesizing the Verilog design using an EDA tool such as Xilinx Vivado, the tool generates an RTL (Register Transfer Level) schematic. This schematic is a graphical representation of the synthesized hardware circuit that implements the ALU functionality.

The RTL schematic shows how the Verilog code has been translated into a network of logic gates, arithmetic blocks, and multiplexers. Key elements visible in the RTL schematic include:

- **Adder/Subtractor Block:** The synthesis tool infers a hardware adder from the + and - operators in the Verilog code. This typically maps to LUT (Look-Up Table) chains or carry-lookahead adder structures on the FPGA.
- **Multiplier Block:** The * operator is synthesized into a dedicated DSP (Digital Signal Processing) block on FPGAs or into a tree of adders and partial products, depending on the target device.
- **Bitwise Logic Gates:** The AND, OR, XOR, and NOT operations are implemented directly as parallel arrays of 2-input logic gates operating on all 5 bits simultaneously.
- **Barrel Shifter:** The left and right shift operations are synthesized into a barrel shifter — a network of multiplexers that can shift a data word by any number of positions in a single clock cycle.
- **Opcode Multiplexer (MUX):** A large 12-to-1 multiplexer selects which operation's output is presented on the result port, controlled by the 4-bit op input. This is the central selector of the ALU.

The synthesized design maps efficiently to FPGA lookup tables (LUTs) and logic resources without any feedback loops, confirming its purely combinational nature. The absence of flip-flops or latches in the schematic verifies that the design is correctly inferred as combinational logic.

7. Simulation Results and Functional Output

7.1 Overview of Simulation

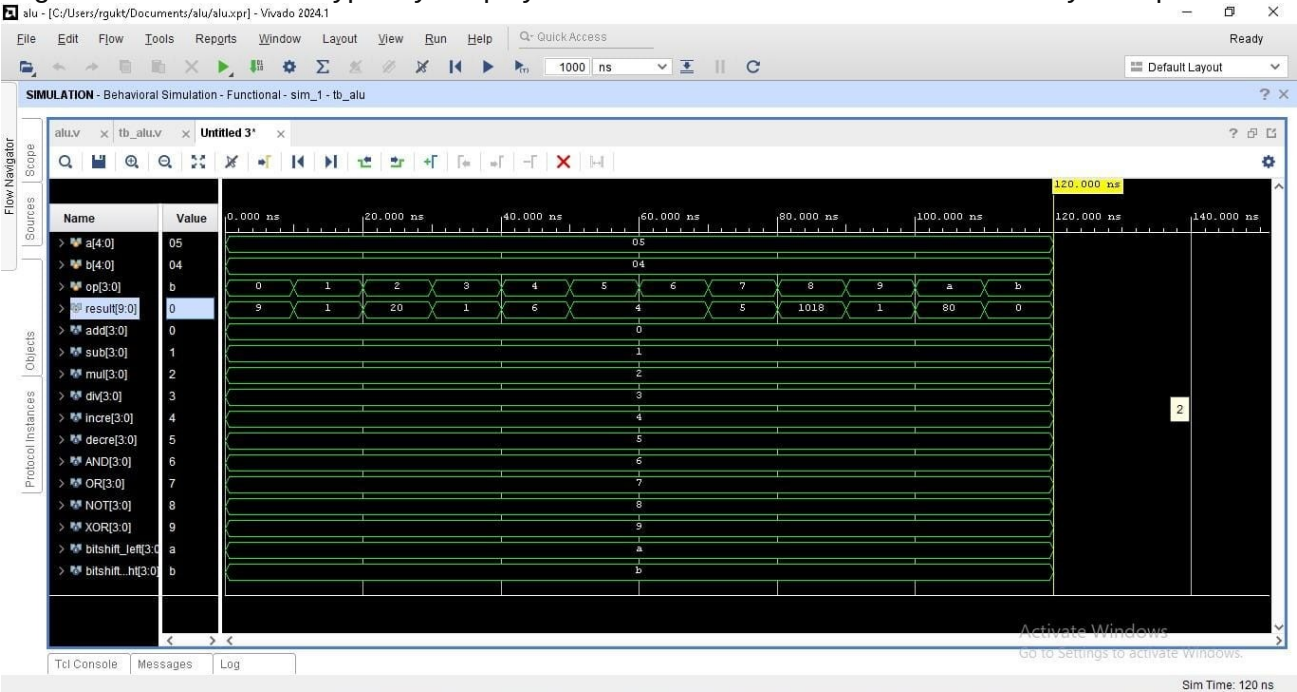
Simulation is the process of running the Verilog design and testbench in a software environment to verify that the hardware behaves as intended before physical implementation. Two industry-standard simulation tools are used in this project: Xilinx Vivado and Mentor ModelSim. Both tools read the Verilog source files, execute the testbench, and generate waveform outputs that can be inspected visually.

The simulation waveforms show the values of all signals (a, b, op, result) plotted over time. Each time the opcode changes (every 10 time units), the result updates to the new computed value. By inspecting the waveform, the engineer can confirm that every operation produces the correct output.

7.2 Vivado Simulation

Xilinx Vivado is a comprehensive FPGA design suite that includes a built-in behavioral simulator (XSim). To run simulation in Vivado, the design and testbench files are added to a project, and the “Run Simulation” option is selected from the Flow Navigator. Vivado compiles the Verilog files, elaborates the design hierarchy, and opens the waveform window showing the testbench signals over time.

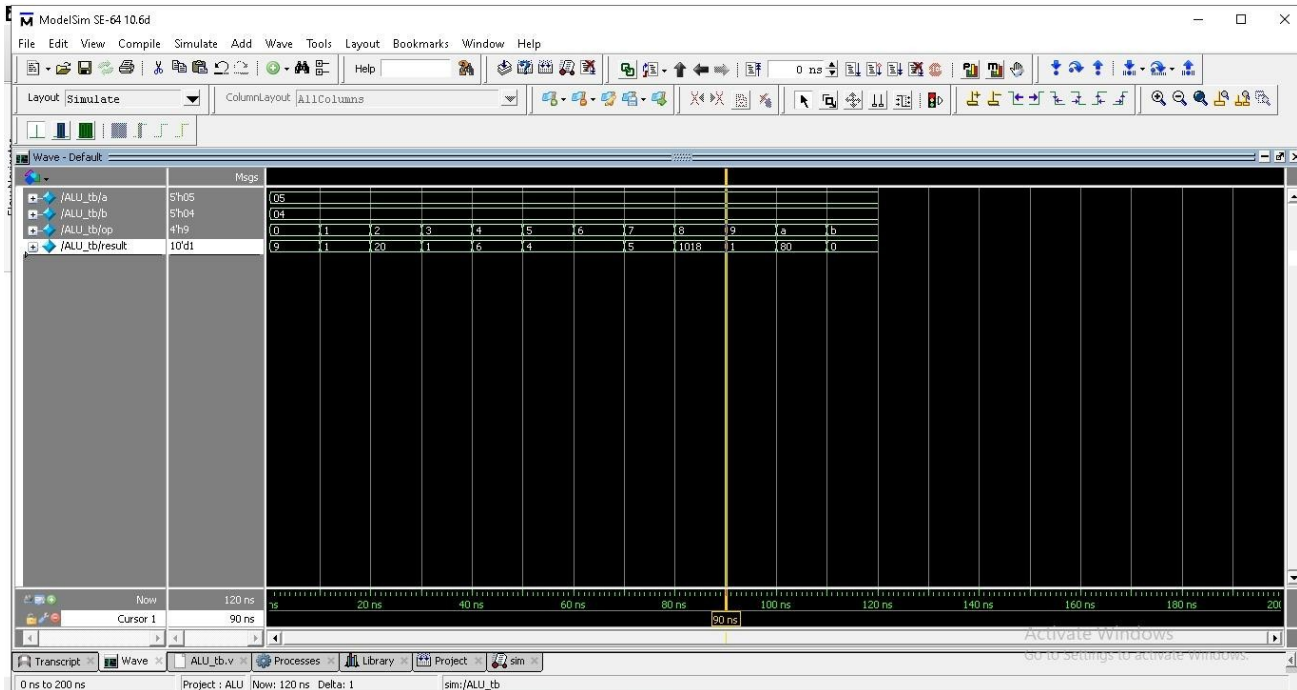
In the Vivado waveform window, each signal is displayed as a horizontal band. Numeric values are shown in the waveform whenever the signal is stable, and transitions are shown as vertical edgeThe result bus is typically displayed in decimal or hexadecimal for easy interpretation



7.3 ModelSim Simulation

Mentor ModelSim (also available as QuestaSim) is a standalone HDL simulation environment widely used in academia and industry. It supports both Verilog and VHDL simulation with a powerful waveform viewer. To simulate in ModelSim, the design and testbench files are compiled using the vlog command, the design is loaded with vsim, and waveforms are added to the Wave window using the add wave command. The simulation is then run for a sufficient number of time steps.

ModelSim provides advanced features such as signal grouping, radix formatting, and cursor-based time measurement, which make it easy to verify that each ALU operation produces the expected result at the correct time. The \$finish statement in the testbench automatically terminates the simulation after all test cases have run.



7.4 Simulation Analysis and Observations

Behavioral simulation confirms correct operation of the ALU under all 12 opcode conditions. The following observations were made from the simulation waveforms:

- Arithmetic operations (add, sub, mul, div, incre, decre) produce numerically correct results for the given operand values (a=5, b=4).
- Logical operations (AND, OR, XOR, NOT) produce correct bitwise outputs matching manual computation.
- Shift operations (bitshift_left, bitshift_right) produce results consistent with the expected power-of-2 multiplication and division.
- The result output updates immediately (zero propagation delay in behavioral simulation) each time the opcode changes, confirming the combinational nature of the design.
- No undefined (X) or high-impedance (Z) values appear on the result output during any valid opcode, confirming the completeness of the case statement coverage.

- The simulation validates output correctness and stable combinational behavior without timing inconsistencies or glitches.

8. Future Scope

This ALU design provides a strong foundation that can be extended in several directions for academic research or industrial application:

1. **Integration with Processor Datapath:** The ALU can be connected to a register file, program counter, and instruction decoder to form a simple RISC-style processor pipeline. This would allow the ALU to execute real assembly instructions, turning this standalone module into the computational core of a complete CPU.
2. **Support for Floating Point Operations:** The current ALU handles only integer arithmetic. Extending it to support IEEE 754 single-precision or double-precision floating point would require dedicated floating-point adder, multiplier, and divider units, significantly expanding the capability of the design.
3. **FPGA Hardware Deployment:** The synthesizable Verilog code can be downloaded onto a physical FPGA board (such as a Xilinx Basys 3 or Nexys A7) using Vivado's programming tools. Hardware switches can provide operand inputs and seven-segment displays can show the result, creating a fully functional physical ALU demonstration.
4. **High-Speed Pipelined ALU Design:** For high-frequency applications, the ALU can be broken into pipeline stages using flip-flops between stages. A pipelined ALU accepts new operands every clock cycle and achieves higher throughput at the cost of one cycle of latency.
5. **Integration with Complete CPU Architecture:** The ALU is the computational heart of a CPU. It can be integrated with a full instruction set architecture (ISA) implementation, including memory access units, branch prediction, and hazard detection logic, to form a complete multi-cycle or pipelined processor.
6. **Power-Optimized ALU for Low-Power Designs:** Techniques such as clock gating, operand isolation, and dynamic voltage scaling can be applied to reduce the power consumption of the ALU, making it suitable for battery-powered embedded systems and IoT devices.

9. Conclusion

This project successfully demonstrates the design, implementation, and verification of a 12-operation Arithmetic Logic Unit using Verilog HDL. The ALU was designed using combinational logic with opcode-based operation selection, making it efficient, modular, and straightforward to understand. The design covers all fundamental categories of digital computation: arithmetic operations, logical operations, and shift operations.

The Verilog source code is clean, synthesizable, and well-structured using localparam constants for readability. The testbench systematically exercises every supported operation and verifies correctness through simulation. Both Vivado and ModelSim simulations confirm that the design produces the expected output for all 12 opcodes with no undefined behavior.

The RTL schematic generated after synthesis reveals how the behavioral Verilog code translates into real hardware elements — adders, multiplexers, logic gates, and shift networks — all correctly

inferred by the synthesis tool. The design maps cleanly to FPGA resources without unintended latches or combinational loops.

This project provides a complete, practical demonstration of digital system design methodology: from behavioral specification through HDL coding, simulation verification, and RTL synthesis. The knowledge and skills developed through this project form the basis for more advanced digital design work, including full processor implementation, FPGA deployment, and custom ASIC development.

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